

SHETH L.U.J. AND SIR M.V. COLLEGE

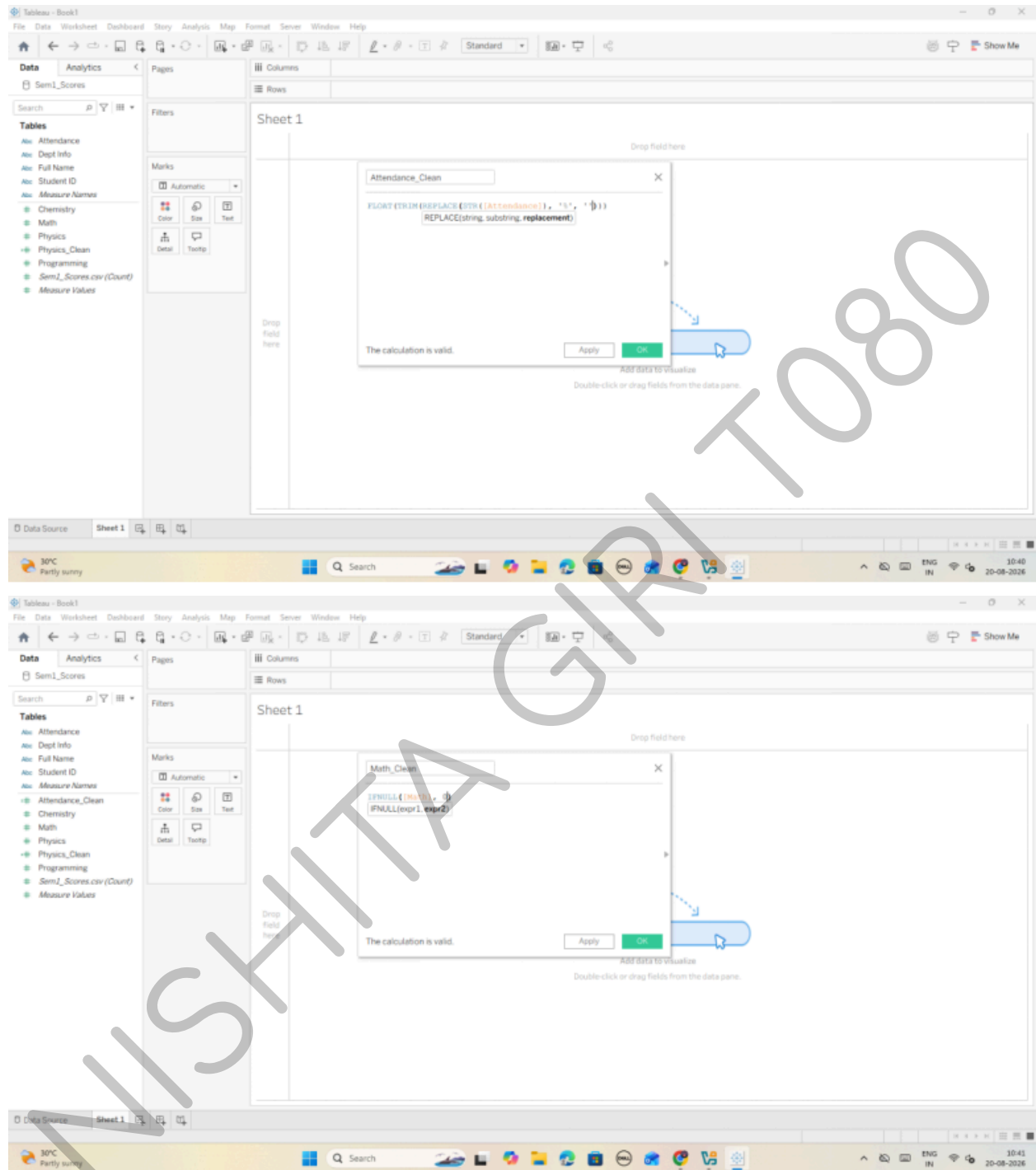
AIM: Cleaning and Structuring Data for Analysis

- Identify and handle missing or inconsistent data
- Convert wide data into tall format

The screenshot displays the Tableau Desktop interface. The top window, titled 'Tableau - Book1', shows the 'Sem1_Scores' data source connected to a text file. The data is presented in a table with 9 fields and 10 rows. The fields are: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The data is organized into columns, with some columns having multiple rows of data. The bottom window, titled 'Tableau - Book1', shows the 'Sem1_Scores' data source in the 'Columns' shelf. A calculation dialog box is open, showing the formula: `FLOR (REPLACE (STR ( [Physics] ), " ", "" ))`. The dialog box also includes a 'REPLACE' function and a 'The calculation is valid' message. The background of the screenshot is watermarked with 'NISHITA GIRI T080'.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance
S301	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %
S302	priya VERMA	ECE-02	null	0.750000	80	85	90 %
S303	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %
S304	Sneha Patel	IT-02	92	0.880000	90	96	95 %
S305	aditya Rao	MECH-04	45	null	50	60	65 %
S306	Aarav Singh	CSE-01	88	0.910000	85	92	88 %
S307	Rahul Nair	ECE-02	null	0.700000	75	80	72 %

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NISHITA GIRI T080
PRACTICAL NO : 9

The top screenshot shows the Tableau Desktop interface with the 'Chemistry_Clean' dialog box open. The dialog box contains the formula `IFNULL([Chemistry], IFNULL(expr1, expr2))`. The 'Add to Visualization' button is highlighted, and the 'The calculation is valid.' message is displayed. The bottom screenshot shows the same interface with the 'Physics_Clean' dialog box open. The dialog box contains the formula `FIRST(ERF(SIN([Physics])), '5', '')`. The 'Add to Visualization' button is highlighted, and the 'The calculation is valid.' message is displayed.

The screenshot displays the Tableau Desktop interface. On the left, the 'Data Source' pane shows the 'Sem1_Scores' table with fields like Student ID, Full Name, Dept Info, and various scores. The 'Marks' card is set to 'Automatic'. The main view shows a calculated field 'Physics_Clean' with the formula: `IFNULL(FLOAT(REPLACE(STR([Physics]), ' ', '')), 0)`. The interface includes a top menu bar, a toolbar, and a status bar at the bottom.

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Text file

Sem1_Scores.csv

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
healthcare-da_roke-data.csv
Indian_Clima_024_2025.csv
iris - iris - iris.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
StressLevelDa_taset (1).csv
StressLevelDa_taset (2).csv
StressLevelDa_etDataset.csv
Student_Activities.csv
Students.csv
supermarket_et_sales.csv

New Union
New Table Extension

Sem1_Scores

10 fields: 40 rows

Name: Sem1_Scores.csv

Description: No description available

Fields

Type	Field Name	Physical Table	Remote Field
Text	Student ID	Sem1_Scores.csv	Student ID
Text	Full Name	Sem1_Scores.csv	Full Name
Text	Dept Info	Sem1_Scores.csv	Dept Info
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Sem1_Scores.csv	Subject
Text	Score	Sem1_Scores.csv	Score
Text	Physics_Ck	Sem1_Scores.csv	Physics_Ck
Text	Attendance_C	Sem1_Scores.csv	Attendance_C
Text	Math_Clean	Sem1_Scores.csv	Math_Clean

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi fact analysis

Need help adding data? Learn more

Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Ck	Attendance_C	Math_Clean	Chems
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Math	0.7800	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Physics	0.7800	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Chemistry	78.0000	null	78.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	

30°C
Partly sunny

Search

ENG IN

10:56
20-08-2026

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Text file

Sem1_Scores.csv

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
healthcare-da_roke-data.csv
Indian_Clima_024_2025.csv
iris - iris - iris.csv
Sem1_Scores.csv
Sem2_Scores.Csv.txt
StressLevelDa_taset (1).csv
StressLevelDa_taset (2).csv
StressLevelDa_etDataset.csv
Student_Activities.csv
Students.csv
supermarket_et_sales.csv

New Union
New Table Extension

Sem1_Scores

10 fields: 40 rows

Name: Sem1_Scores.csv

Description: No description available

Fields

Type	Field Name	Physical Table	Remote Field
Text	Student ID	Sem1_Scores.csv	Student ID
Text	Full Name	Sem1_Scores.csv	Full Name
Text	Dept Info	Sem1_Scores.csv	Dept Info
Text	Attendance	Sem1_Scores.csv	Attendance
Text	Subject	Sem1_Scores.csv	Subject
Text	Score	Sem1_Scores.csv	Score
Text	Physics_Ck	Sem1_Scores.csv	Physics_Ck
Text	Attendance_C	Sem1_Scores.csv	Attendance_C
Text	Math_Clean	Sem1_Scores.csv	Math_Clean

Relate tables
Drag tables near Sem1_Scores.csv to relate them

Add a base table
Drag out additional base tables for multi fact analysis

Need help adding data? Learn more

Student ID	Full Name	Dept Info	Attendance	Subject	Score	Physics_Ck	Attendance_C	Math_Clean	Chems
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.0000	null	
S001	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.0000	null	
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Math	0.7800	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Physics	0.7800	null	90.0000	null	
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Chemistry	78.0000	null	78.0000	null	
S003	Rohan Gupta	CSE-01	78 %	Math	78.0000	null	78.0000	null	

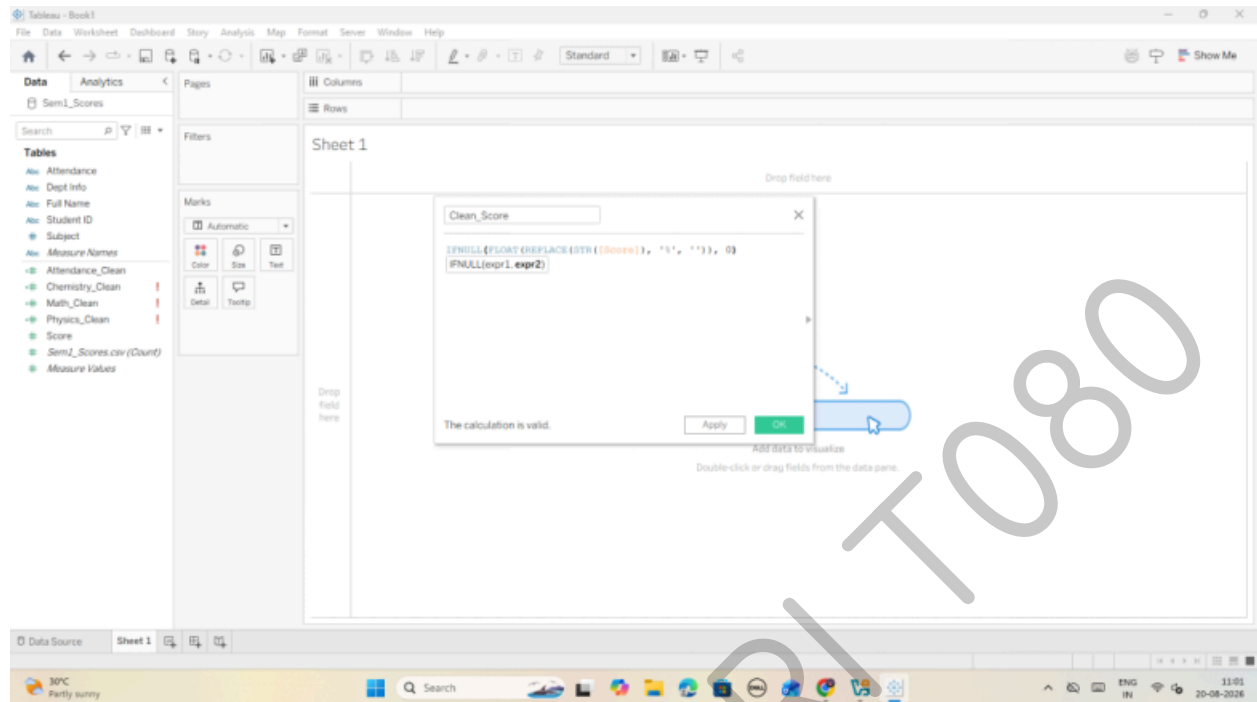
30°C
Partly sunny

Search

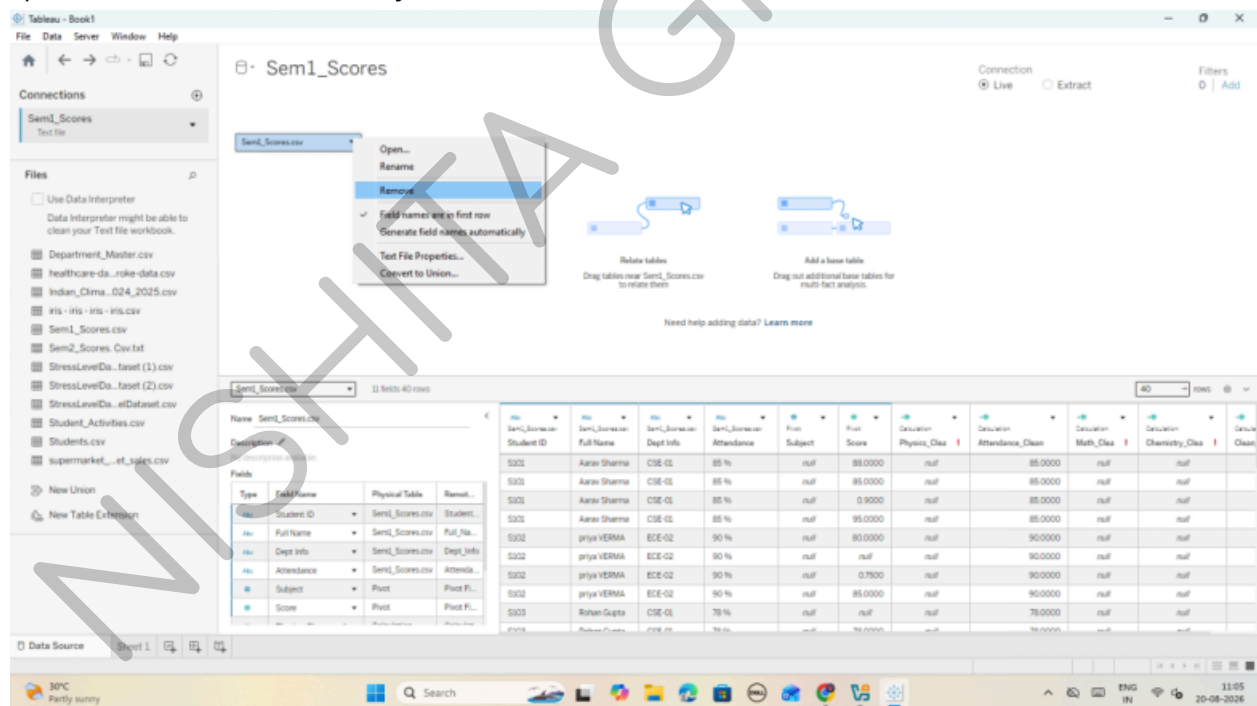
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- c) Perform union of multiple files
- d) Execute cross-database joins



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The screenshot displays the Tableau Desktop interface. The 'Connections' pane on the left shows 'Sem1_Scores' as a text file. The 'Files' pane lists various CSV files. The 'Union' dialog box is open, showing 'Sem1_Scores.csv' and 'Sem2_Scores.csv' as tables in the union. The 'Data Source' pane shows 'Sem1_Scores' as a text file. The 'Fields' pane shows the fields for 'Sem1_Scores.csv'.

Union Dialog Box:

- Connection: Sem1_Scores
- Tables in union: 2
- Buttons: Apply, OK

Data Source:

- Connection: Live
- Extract: Extract
- Filters: 0 | Add

Fields:

Type	Field Name	Physical Table	Remote Table
Text	Dept info	Sem1_Scores.csv	Dept info
Text	Math	Sem1_Scores.csv	Math
Text	Physics	Sem1_Scores.csv	Physics
Text	Chemistry	Sem1_Scores.csv	Chemistry
Text	Programming	Sem1_Scores.csv	Programming

Table View:

Student ID	Full Name	Dept info	Math	Physics	Chemistry	Programming	Attendance	PHYSICS_CLEAN
S001	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	0.900000
S002	priya VERMA	ECE-02	null	0.750000	80	85	90 %	0.750000
S003	Rohan Gupta	CSE-03	78	0.820000	null	88	78 %	0.820000
S004	Sneha Patel	IT-03	92	0.880000	90	96	95 %	0.880000
S005	aditya Rao	Mech-04	45	null	50	60	65 %	0.000000
S006	Ananya Singh	CSE-01	88	0.980000	85	92	88 %	0.980000
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	0.700000
S008	Kavya Reddy	IT-03	95	0.940000	92	98	98 %	0.940000

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PRACTICAL NO : 9

The screenshot displays the Tableau Desktop interface for a worksheet named 'Sem1_Scores'. The left sidebar shows the 'Connections' pane with 'Sem1_Scores' selected as a text file, and the 'Files' pane listing various CSV files. The main workspace shows a data table with columns: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, Attendance, and Calculation. The table contains 10 rows of data. A large 'UNTHA' watermark is overlaid on the center of the image.

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Sem1_Scores

Connections: Sem1_Scores (Text file)

Files: Sem1_Scores.csv, Sem2_Scores.Csv.txt, Sem2_Scores.csv, startup_funding.csv, store_sales_data (2).csv, StressLevel...set (1).csv, StressLevel_Dataset.csv, Student_Activities.csv, Student_Performance.csv, Students.csv, supermarket_sales (2).csv, supermarket_sales.csv, table-1785...716728.csv, table-1785...987032.csv, Titanic-Dataset.csv, Titanic.csv, weatherHistory.csv

Fields:

Type	Field Name	Physical Table	Remote...
Text	Attendance	Sem1_Scores.csv	Attendanc...
Text	Subject	Pivot	Pivot Fel...
Text	Score	Pivot	Pivot Fel...
Text	PHYSICS_CLEAN	Calculation	Calculati...
Text	Attendance_clean	Calculation	Calculati...

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Student ID	Full Name	Dept Info	Attendance	Subject	Score	PHYSICS_CLEAN	Attendance_clean
S001	Aarav Sharma	CSE-01	85 %	Chemistry	88.0000	null	85.00
S002	Aarav Sharma	CSE-01	85 %	Math	85.0000	null	85.00
S003	Aarav Sharma	CSE-01	85 %	Physics	0.9000	null	85.00
S004	Aarav Sharma	CSE-01	85 %	Programming	95.0000	null	85.00
S002	priya VERMA	ECE-02	90 %	Chemistry	80.0000	null	90.00
S002	priya VERMA	ECE-02	90 %	Math	null	null	90.00
S002	priya VERMA	ECE-02	90 %	Physics	0.7500	null	90.00
S002	priya VERMA	ECE-02	90 %	Programming	85.0000	null	90.00

[illegible]

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The top screenshot shows the 'Union' dialog box in Tableau. The 'Specific (manual)' tab is selected, and the 'Connection: Sem1_Scores' is set. The files 'Sem1_Scores.csv' and 'Sem2_Scores.csv' are listed in the 'Tables in union: 2' section. The 'Apply' button is highlighted.

The bottom screenshot shows the Tableau interface after the union has been created. The 'Union' table is visible in the 'Data Source' pane. The 'Fields' pane shows the fields from both tables, including 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Chemistry', 'Programming', 'Attendance', and 'Table Name'. The 'Columns' shelf contains the fields 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Chemistry', 'Programming', 'Attendance', and 'Table Name'.

Type	Field Name	Physical Table	Remote F...
Ali	Student_ID	Union	Student_ID
Ali	Full_Name	Union	Full_Name
Ali	Dept_Info	Union	Dept_Info
Ali	Math	Union	Math
Ali	Physics	Union	Physics

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The top screenshot shows the Tableau interface with a calculated field dialog box open. The dialog contains the following logic:

```
IF CONTAINS([Table Name], "Sem1")
THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2")
THEN "Semester 2" ELSE "Other"
END
```

The bottom screenshot shows the 'Sem1_Scores' data source configuration. It displays a union of 'Sem1_Scores' and 'Department_Master.csv'. Below this, a table of data is shown:

Student_ID	Full Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S001	Aarav Sharma	CSE-01	85	0.900000	88	95	85 %	Sem1_Scores.csv
S002	priya VERMA	ECE-02	null	0.750000	80	85	90 %	Sem1_Scores.csv
S003	Rohan Gupta	CSE-01	78	0.820000	null	88	78 %	Sem1_Scores.csv
S004	Sreha Patel	IT-03	92	0.880000	90	96	95 %	Sem1_Scores.csv
S005	aditya Rao	MECH-04	45	null	50	60	65 %	Sem1_Scores.csv
S006	Ananya Singh	CSE-01	88	0.960000	85	92	88 %	Sem1_Scores.csv
S007	Rahul Nair	ECE-02	null	0.700000	75	80	72 %	Sem1_Scores.csv
S008	Kavya Reddy	IT-03	95	0.940000	92	98	96 %	Sem1_Scores.csv

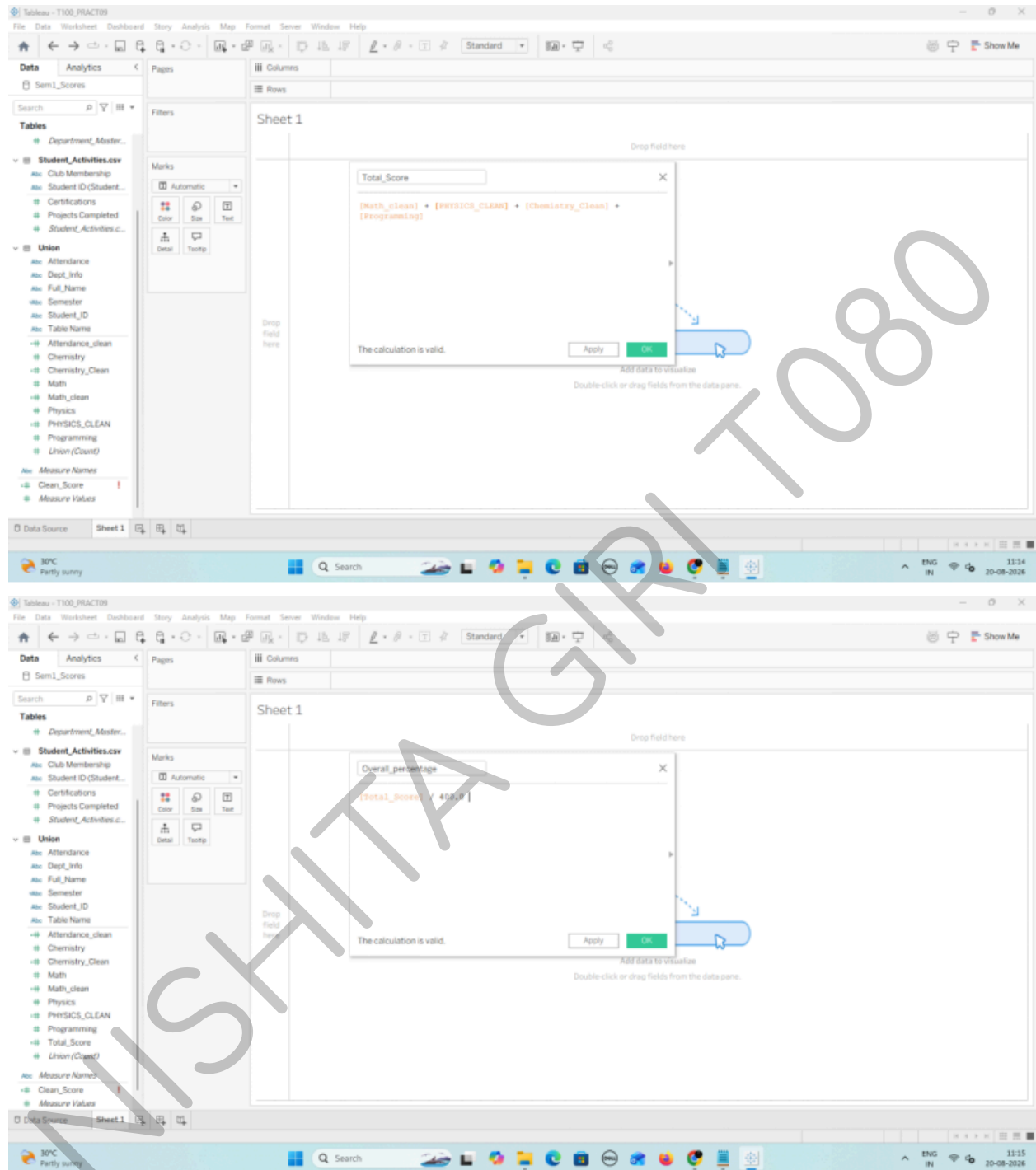
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The top screenshot shows the Tableau interface with a data source named 'Sem1_Scores' connected to a text file. A union relationship is established between 'Department_Master.csv' and 'Student_Activities.csv'. The data preview shows a table with columns: Student ID (Student Activities), Club Membership, Projects Completed, and Certifications. The bottom screenshot shows a blank Tableau worksheet named 'Sheet 1' with a data source of 'Sem1_Scores'. The data pane on the left lists fields from 'Department_Master.csv', 'Student_Activities.csv', and the 'Union' relationship. The main workspace is empty, with a prompt to 'Add data to visualize'.

Student ID (Student Activities)	Club Membership	Projects Completed	Certifications
S001	Coding Club	3	2
S002	Robotics Club	2	1
S003	Coding Club	2	1
S004	Debate Society	4	3
S005	Sports Club	1	0
S006	Coding Club	3	2
S007	Robotics Club	2	1
S008	Debate Society	4	3

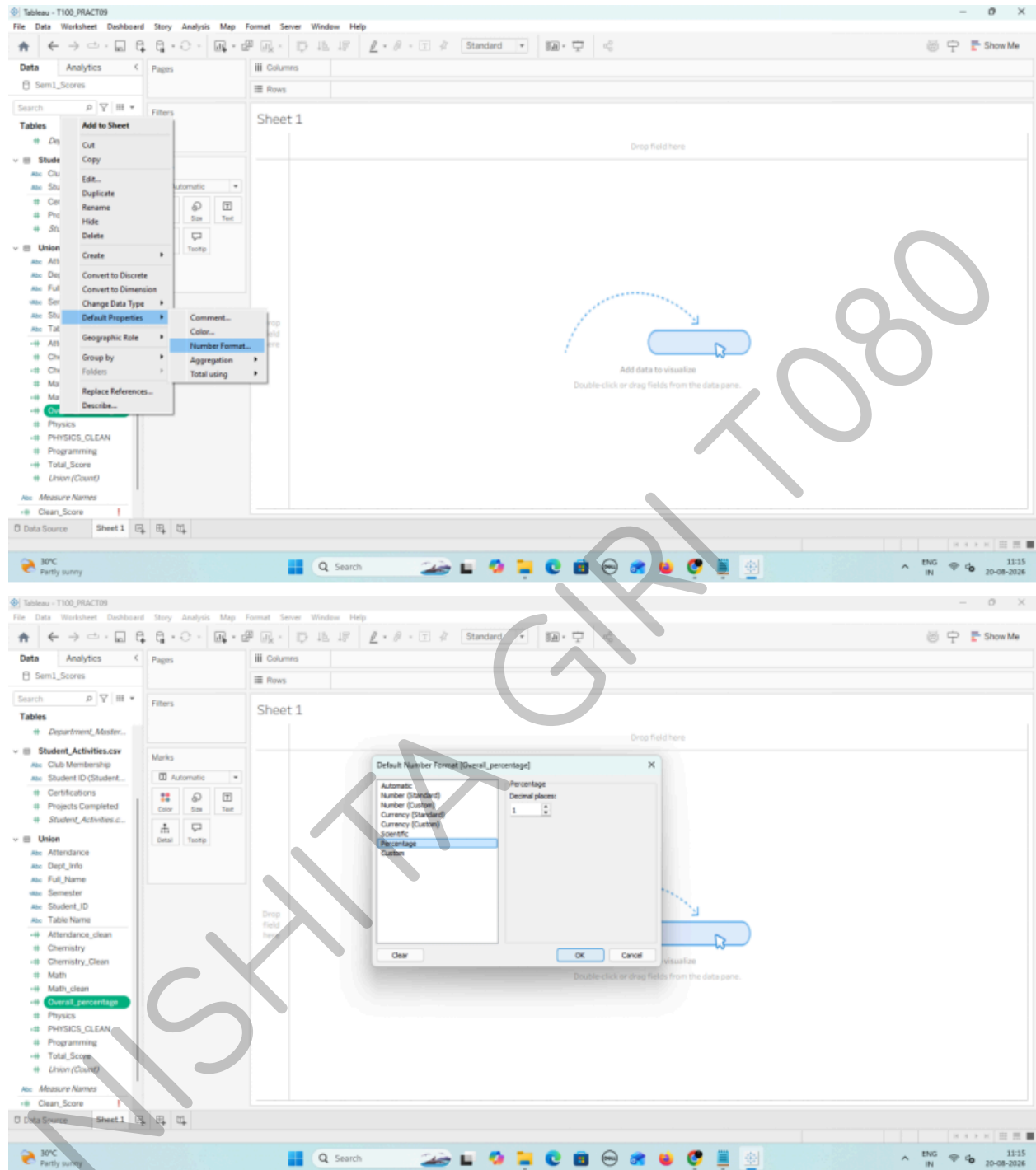
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets

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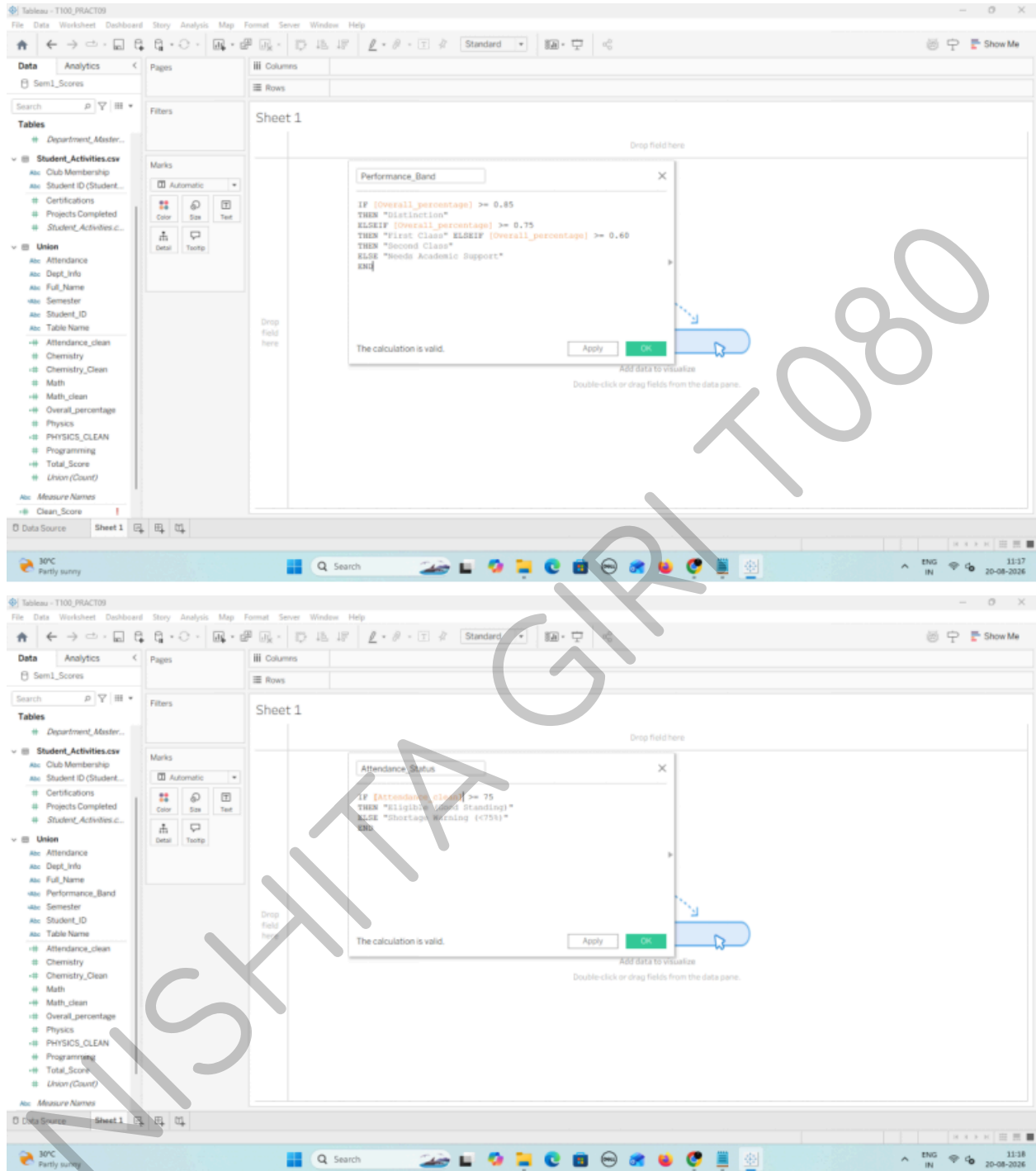


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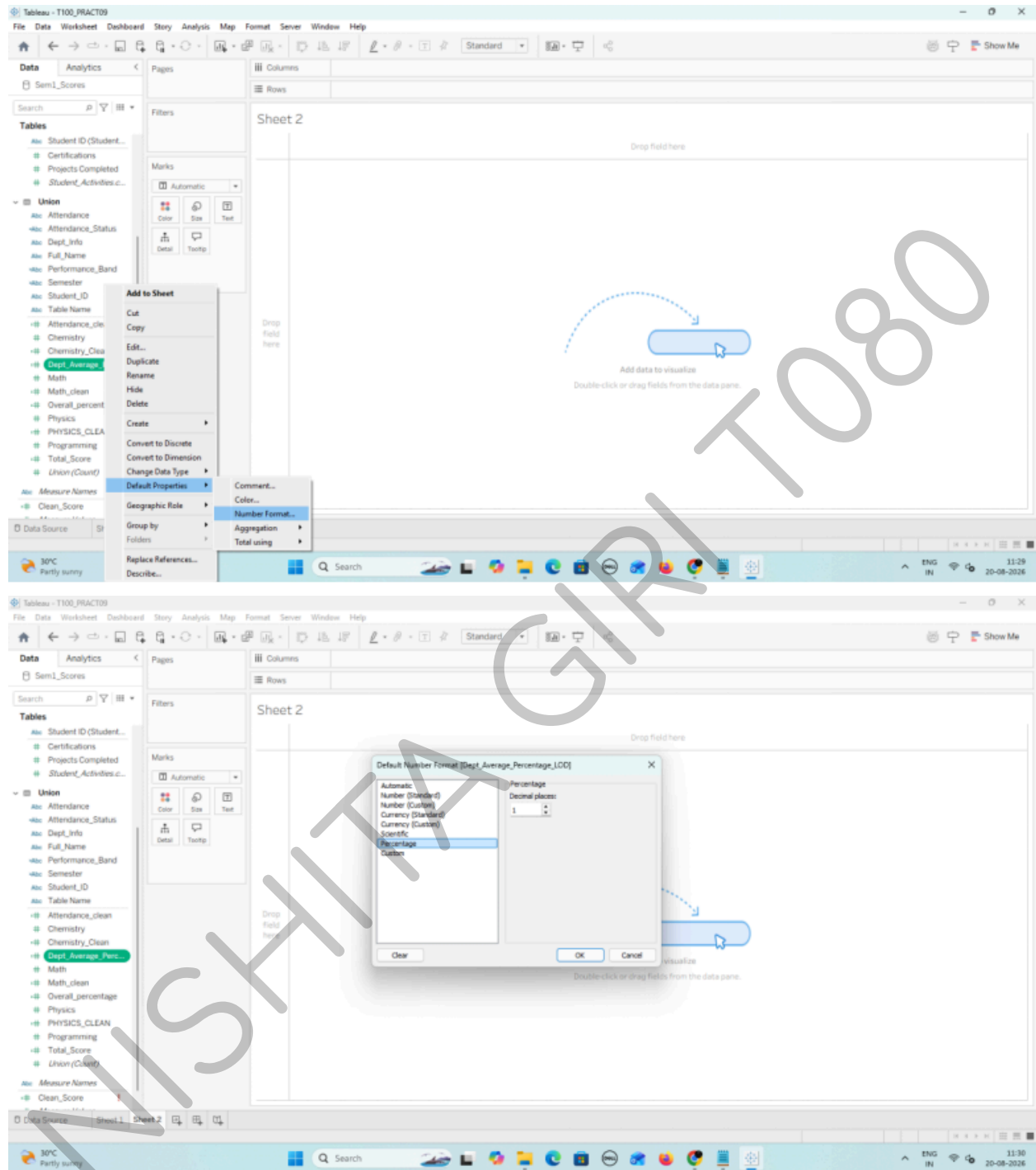
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The top screenshot shows a Tableau worksheet named 'Sheet 1'. The columns are 'Student_ID' and 'Full_Name'. The rows are 'Semester'. The data is a pivot table showing scores for various subjects (S101 to S110) for two semesters. A 'Performance_Band' is calculated based on the 'Overall_Percentage'.

	S101	S102	S103	S104	S105	S106	S107	S108	S109	S110
Semester 1	67.2%	62.4%	41.4%	69.7%	38.8%	66.5%	38.9%	71.5%	49.4%	63.9%
Semester 2	69.2%	62.4%	63.2%	72.0%	43.1%	68.5%	58.7%	72.7%	51.9%	65.5%

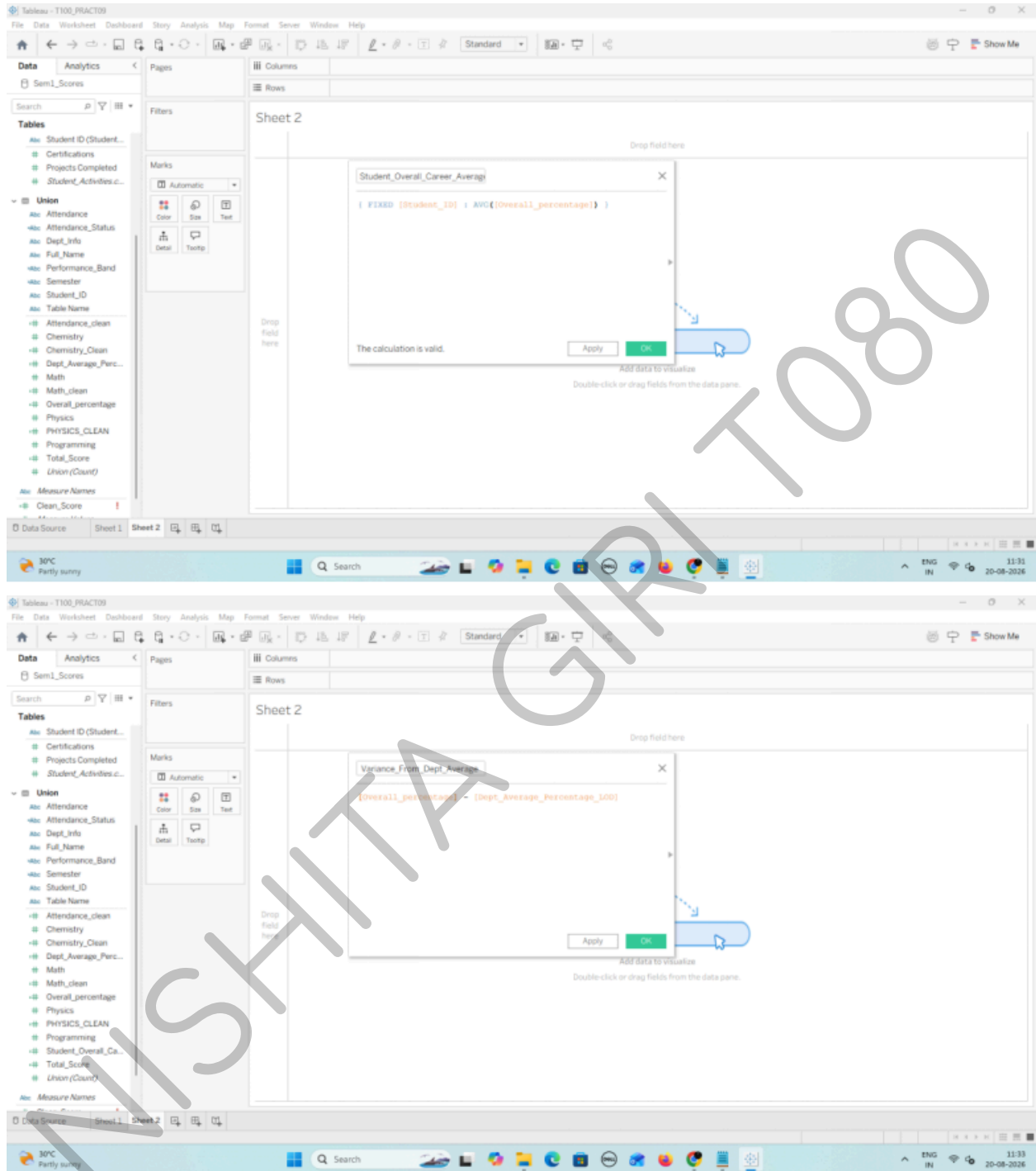
The bottom screenshot shows a Tableau worksheet named 'Sheet 2'. A calculated field is being created: `DEPT_AVERAGE_PERCENTAGE (GO)` with the formula `FIXED (Dept, Info) : AVG (Overall_Percentage) )`. The calculation is valid, and the user is prompted to 'Apply' or 'OK'.

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The first screenshot shows the Tableau interface with a calculated field dialog box open. The dialog contains the following logic:

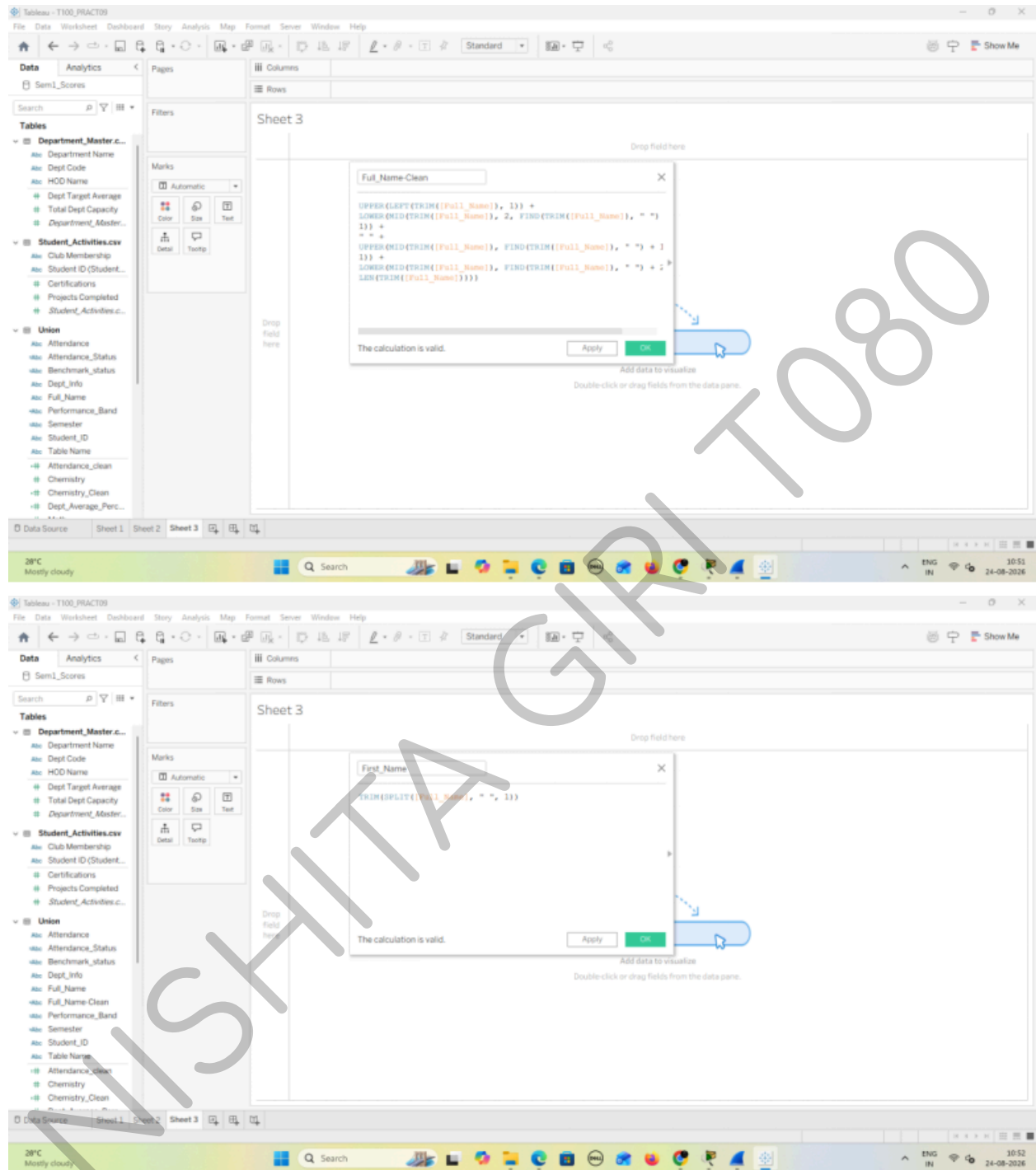
```
IF [Variance_From_Dept_Average] >= 0  
THEN "Above Department Benchmark"  
ELSE "Below Department Benchmark"  
END
```

The second screenshot shows the final visualization in a table format. The columns are Dept_Info, Student_ID, Full_Name, and Semester. The data is grouped by Department (CSE-01, ECE-02, IT-03, MECH-04) and then by Student. The 'Semester' column is split into two columns: 'r1' and 'r2'. The 'BenchMark_status' column is color-coded: blue for 'Above Department B.' and orange for 'Below Department B.'.

Dept_Info	Student_ID	Full_Name	Semester		BenchMark_status
			r1	r2	
CSE-01	S101	Aarav Sharma	67.2%	62.2%	Above Department B.
	S103	Shruti Gupta	41.7%	74.2%	Above Department B.
	S106	Ananya Singh	66.5%	68.5%	Above Department B.
	S109	Vikram Joshi	66.9%	51.9%	Above Department B.
ECE-02	S102	Kavya Verma	64.4%	62.4%	Above Department B.
	S107	Rahul Bhat	60.4%	58.7%	Above Department B.
	S110	Pooja Patel	63.0%	65.5%	Above Department B.
	S104	Snigdha Patil	69.7%	72.0%	Above Department B.
IT-03	S108	Kavya Reddy	71.5%	72.7%	Above Department B.
	S105	Aditya Rao	68.8%	43.1%	Below Department B.

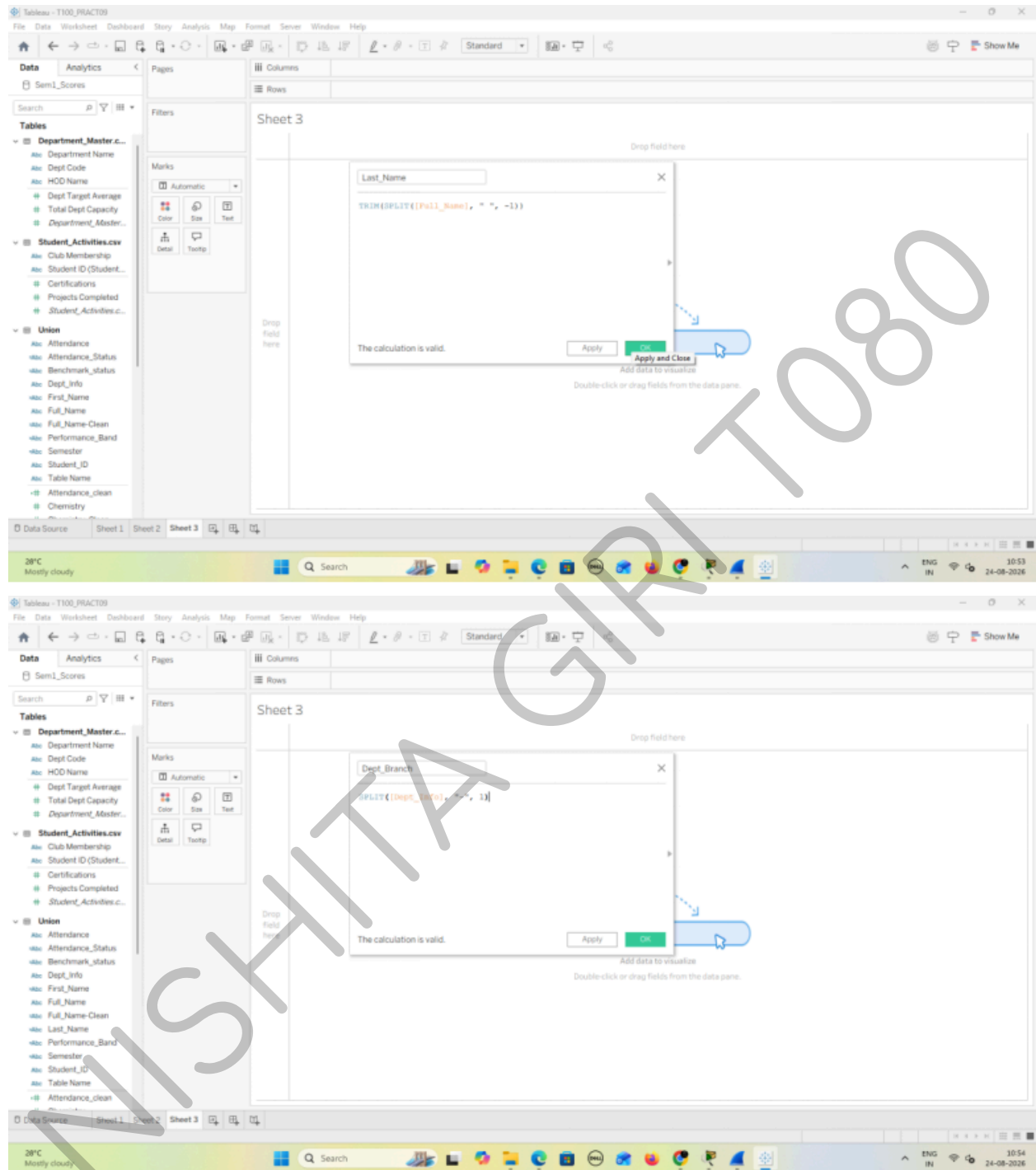
g) Apply advanced data cleaning techniques

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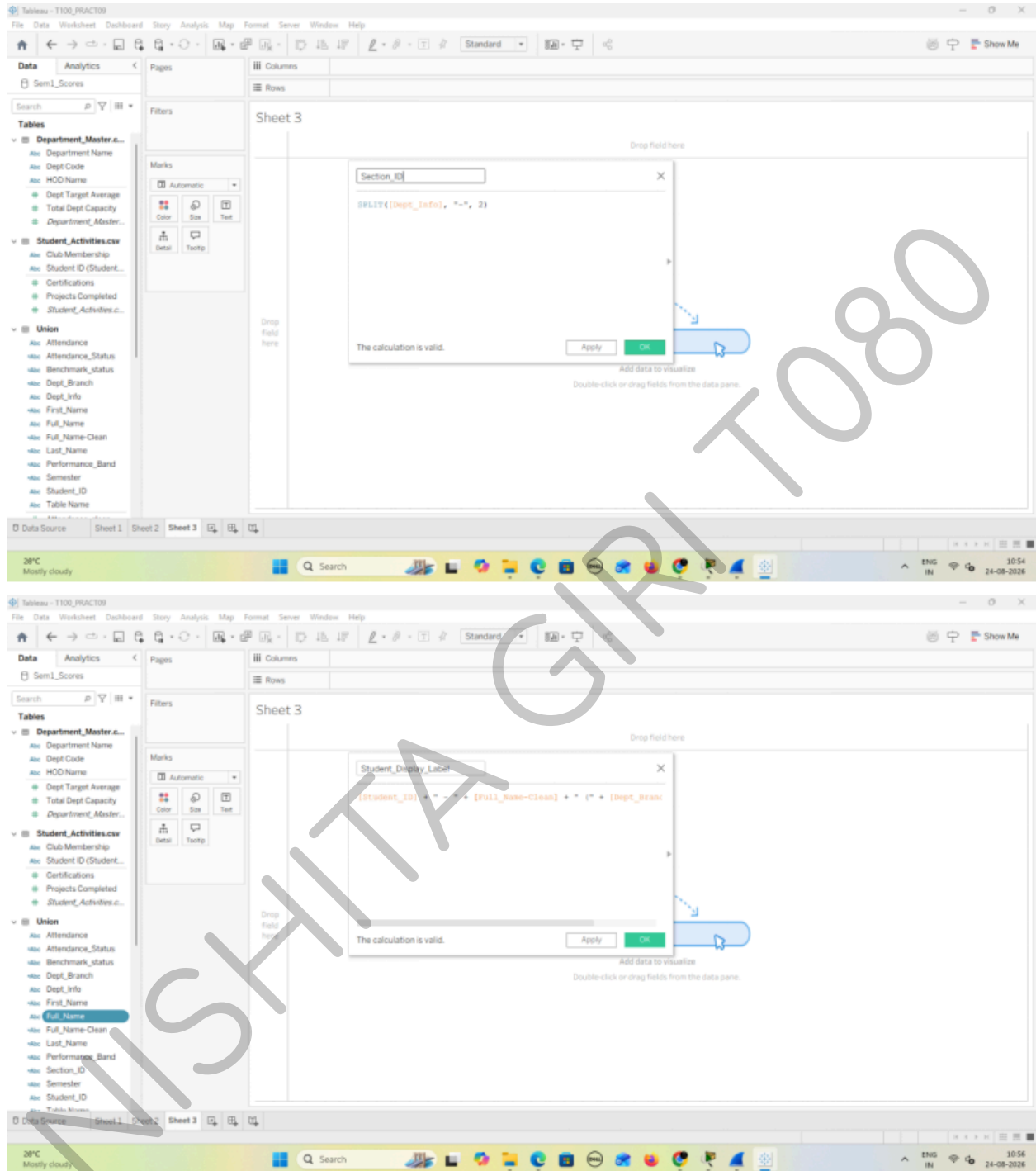
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